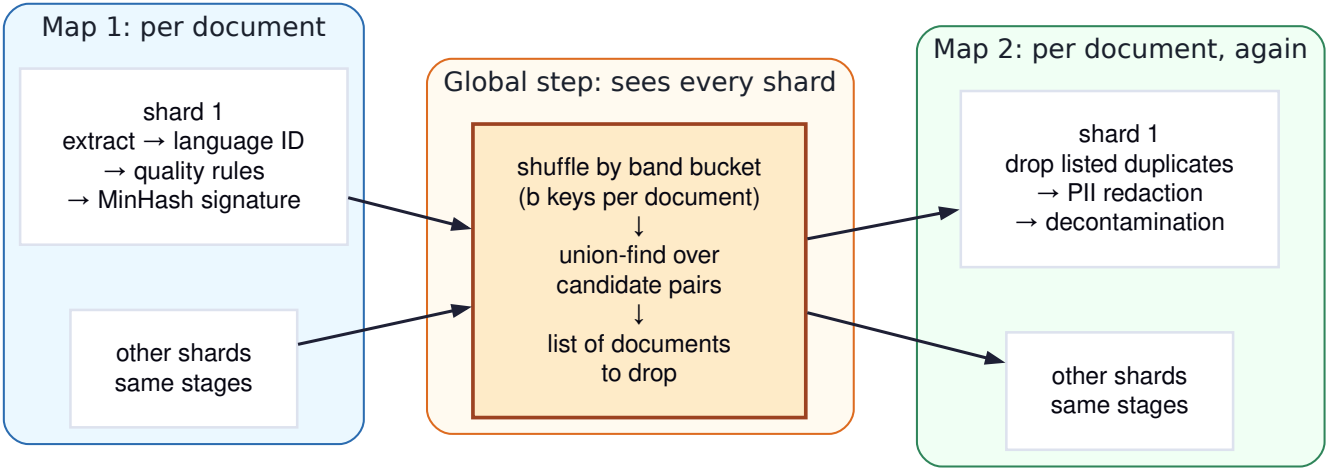
Map 1: per document
shard 1
extract → language ID
→ quality rules
→ MinHash signature
other shards
same stages
Global step: sees every shard
shuffle by band bucket
(b keys per document)
↓
union-find over
candidate pairs
↓
list of documents
to drop
Map 2: per document, again
shard 1
drop listed duplicates
→ PII redaction
→ decontamination
other shards
same stages